TPMS Structures for prosthetic implants

Stiffness grading optimization

SAPIENZA
UNIVERSITÀ DI ROMA

10/04/2026 Dr. Gianluca Santesarti

Outline

- **Stiffness grading optimization**

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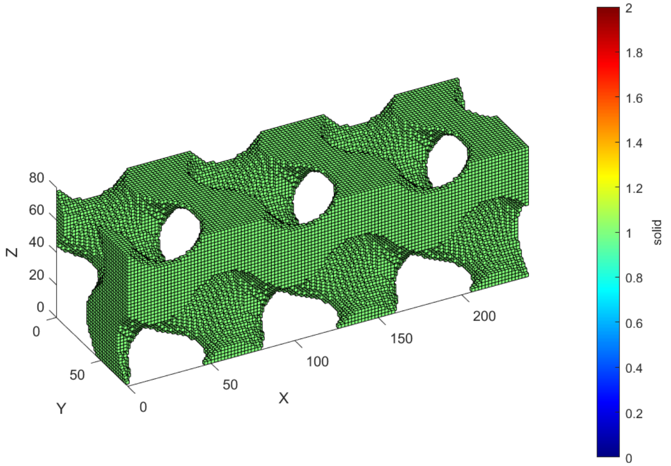
Stiffness grading optimization

x-direction

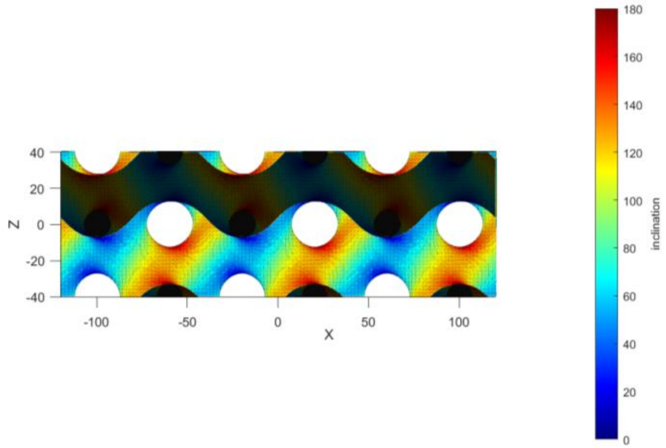
Schoen-Gyroid
$$U_G = \cos(2\pi x)\sin(2\pi y) + \cos(2\pi y)\sin(2\pi z) + \cos(2\pi z)\sin(2\pi x)$$

Example
$$U_G = c \qquad c = 0.3$$

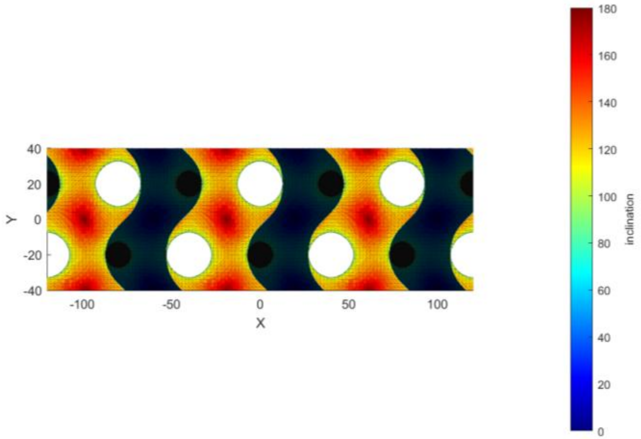
Voxel 3D plot



Surfmesh 3D plot – XZ plane



Surfmesh 3D plot – XY plane

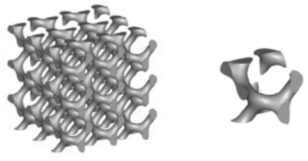


Stiffness grading optimization

x-direction

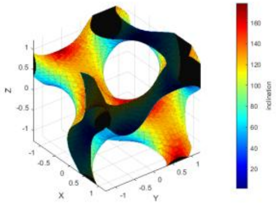
Reference geometry (Naghavi2022)

TPMS: Schoen-Gyroid
Type: Skeletal
Unit cell size: 2.5 mm
Material: Ti-6Al-4V

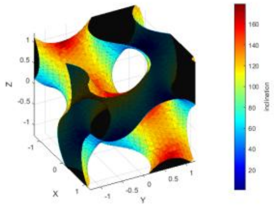


(c) Gyroid

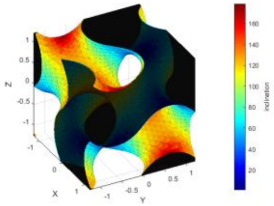
$c = 0.74$



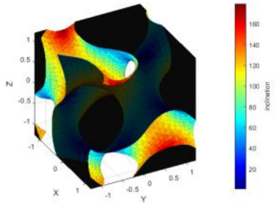
$c = 0.30$



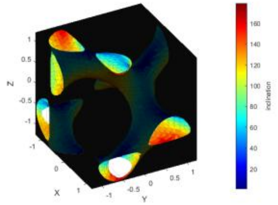
$c = -0.15$



$c = -0.63$



$c = -1.05$



Isovalue-Relative density characterization

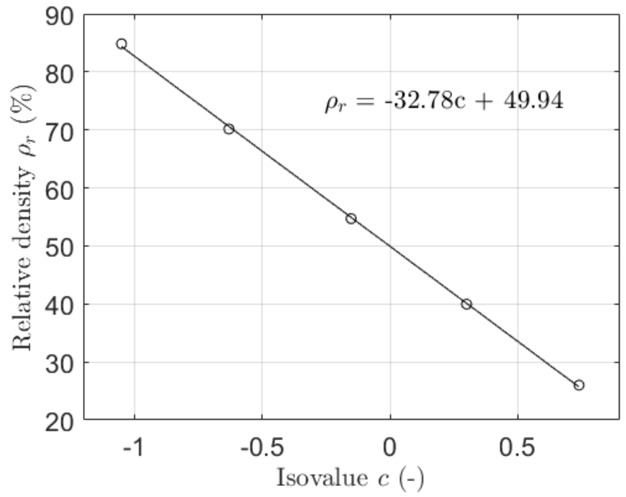
Isovalue range: $c = (-1.05, 0.74)$

Rel. density range: $\rho_r = 26\% - 84.8\%$

Porosity range: $P = 74\% - 15.2\%$

Ref. from trabecular to cortical bone (McGregor2021)

Linear fitting $\rho_r(c) = a * c + b$



(1) *Naghavi et al. (2022). TPMS-based gyroid and diamond Ti6Al4V scaffolds for bone implants. Bioengineering, 9(10), 504.*
(2) *McGregor et al. (2021) Bone parameters titanium lattice design for powder bed fusion additive manufacturing. Additive Manufacturing 47: 102273.*

Stiffness grading optimization

x-direction

Reference geometry

TPMS: Schoen-Gyroid
Type: Skeletal
Unit cell size: 2.5 mm
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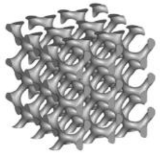


(c) Gyroid

Finit Element Analysis (FEA)

Material: Ti-6Al-4V
Force: 500-2500 N (compression), x direction top
Constraint: fixed bottom

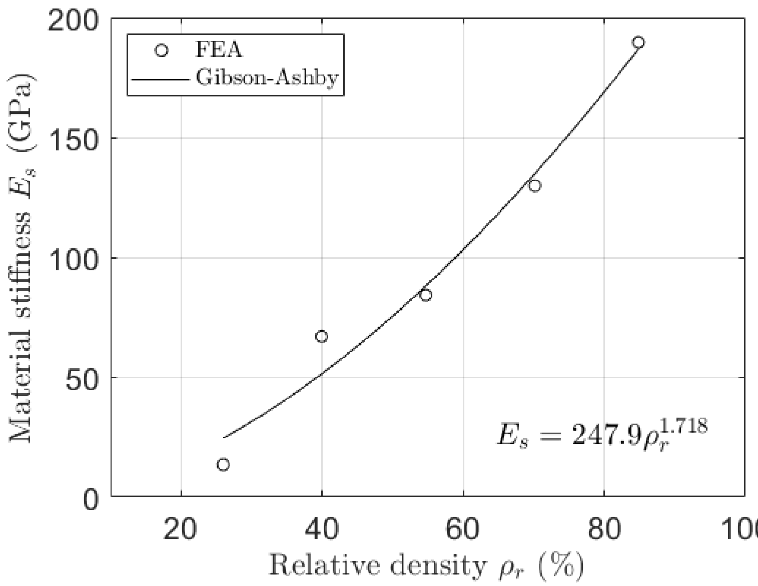
c (-)	ρ_r (%)	E_s (GPa)
0.74	26	13.4*
0.30	40	66.8*
-0.15	54.7	84.1*
-0.63	70.2	129.8*
-1.05	84.8	189.6

from Rometta S. MSc thesis work

Gibson-Ashby model - Fitting

$$E_s(\rho_r) = K\rho_r^n$$

$K = 247.9$ GPa
$n = 1.718$



Stiffness grading optimization

x-direction

Stiffness optimization

$$\boxed{E_{opt}(x) = \alpha x + \beta}$$

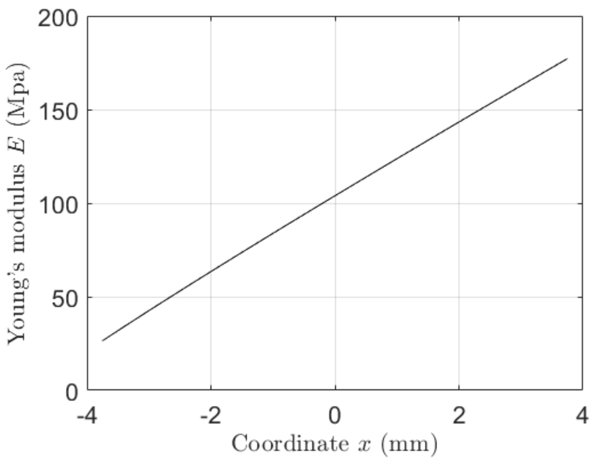
$$\rho_r(c) = a * c + b$$

$$E(\rho_r) = K\rho_r^n$$

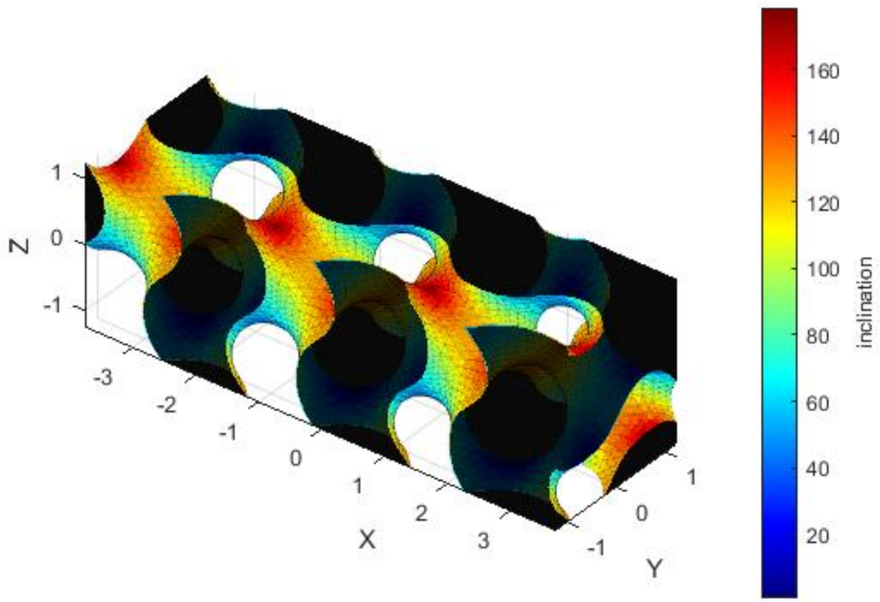
$$\downarrow$$

$$\boxed{\rho_{r,opt}(x) = \left(\frac{\alpha x + \beta}{K}\right)^{1/n}} \qquad \boxed{c_{opt}(x) = \frac{1}{a}\left(\frac{\alpha x + \beta}{K}\right)^{1/n} - \frac{b}{a}}$$

Stiffness target design



Optimized structure (3x1x1 cells)

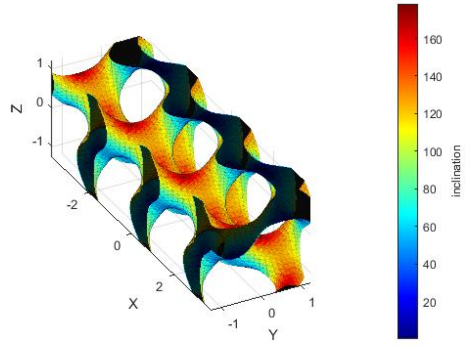


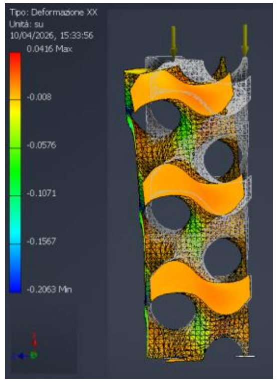
Stiffness grading optimization

x-direction

Comparison

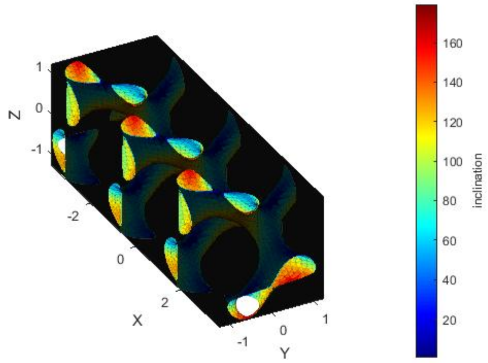
Structure with 74% Porosity (3x1x1 cells)





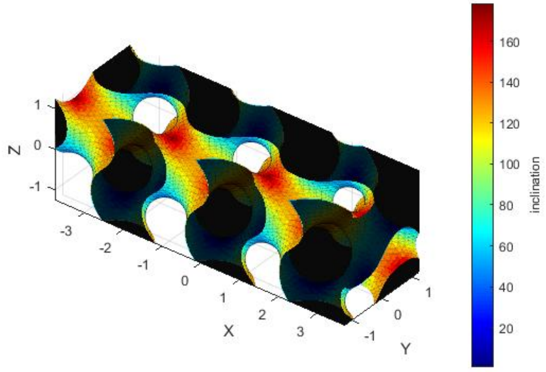
$E_s = 6.75$ GPa

Structure with 15.2% Porosity (3x1x1 cells)



Inventor software error

Optimized structure (3x1x1 cells)



Inventor software error

Thanks for your attention!

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Update 09-2-2026 - Outline

- **Stiffness grading optimization**

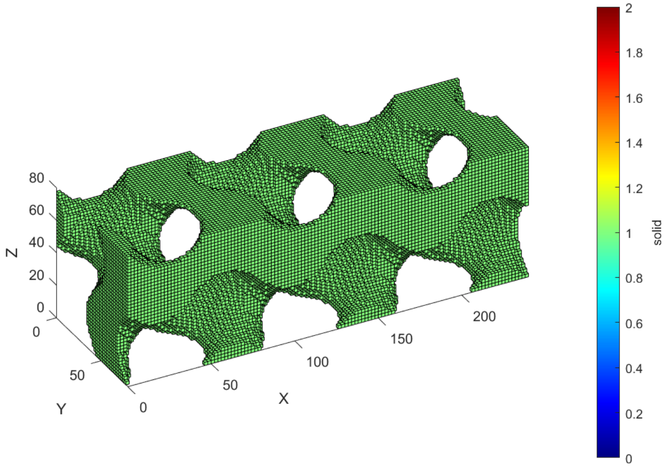
Software

Stiffness grading optimization – x-direction

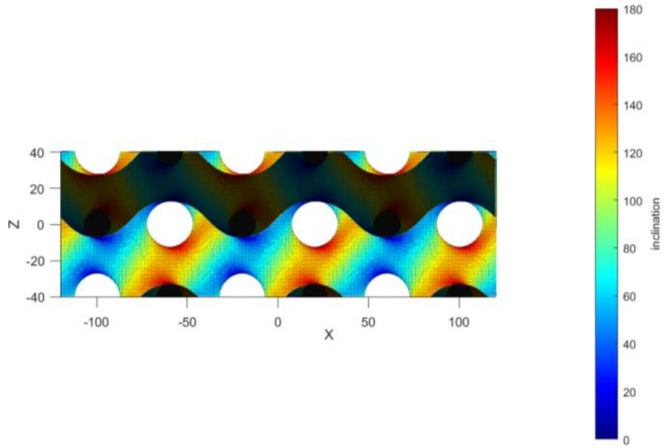
Schoen-Gyroid $$U_G = \cos(2\pi x)\sin(2\pi y) + \cos(2\pi y)\sin(2\pi z) + \cos(2\pi z)\sin(2\pi x)$$

Example $$U_G = c \qquad c = 0.3$$

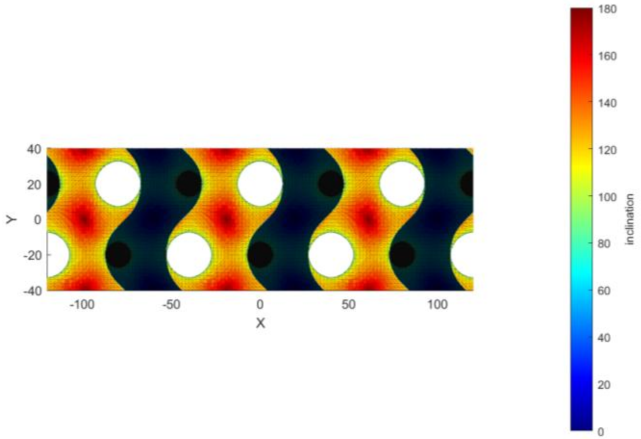
Voxel 3D plot



Surfmesh 3D plot – XZ plane



Surfmesh 3D plot – XY plane

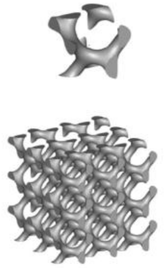


Software

Stiffness grading optimization – x-direction

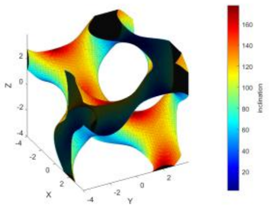
Reference geometry (Abou-Ali2019)

TPMS: Schoen-Gyroid
Unit cell size: 8 mm
Pattern: 5x5x5 cells
Material: polymeric polyamide
Fabrication: laser powder bed fusion-EOS Formiga P110

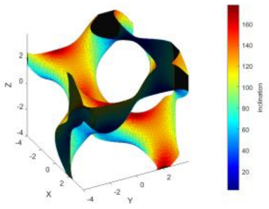


(c) Gyroid

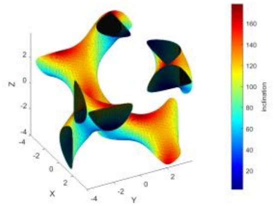
$c = 0.8$



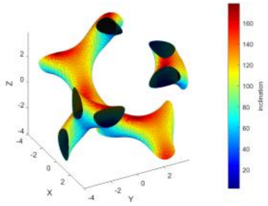
$c = 0.92$



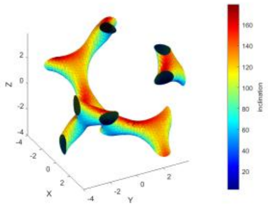
$c = 1.04$



$c = 1.16$



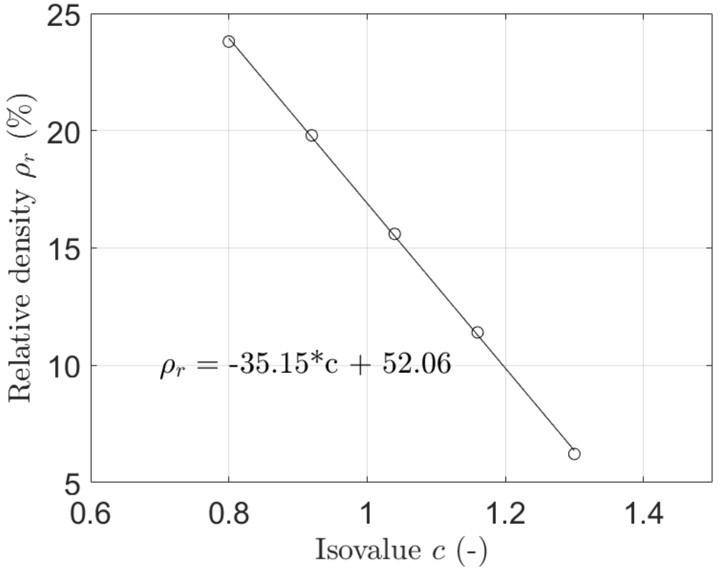
$c = 1.3$



Isovalue-Relative density characterization

Isovalue range: $c = 0.8 - 1.3$
Rel. density range: $\rho_r = 6.1\% - 23.9\%$

Linear fitting $\rho_r(c) = a * c + b$



Abou-Ali2019-Mechanical Response of 3D Printed Bending-Dominated, Journal of Materials Engineering and Performance, 28:2316–2326

Software

Stiffness grading optimization – x-direction

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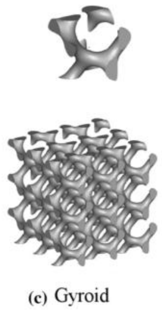
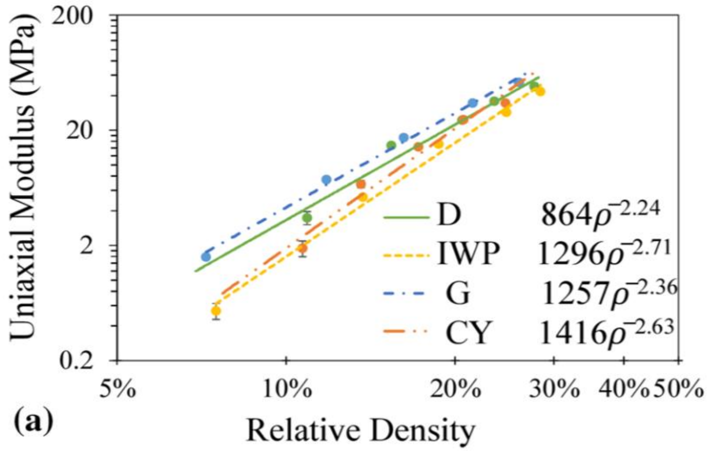
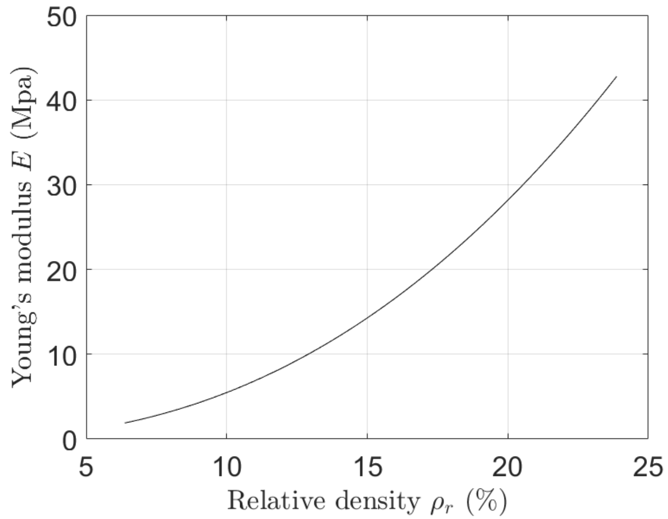


Gibson-Ashby model

$$E(\rho_r) = K\rho_r^n$$

$$K = 1257 \text{ MPa}$$
$$n = 2.36$$





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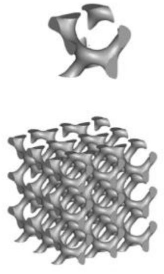

(c) Gyroid

Stiffness optimization

$$\rho_r(c) = a * c + b$$

$$\boxed{E_{opt}(x) = \alpha x + \beta}$$

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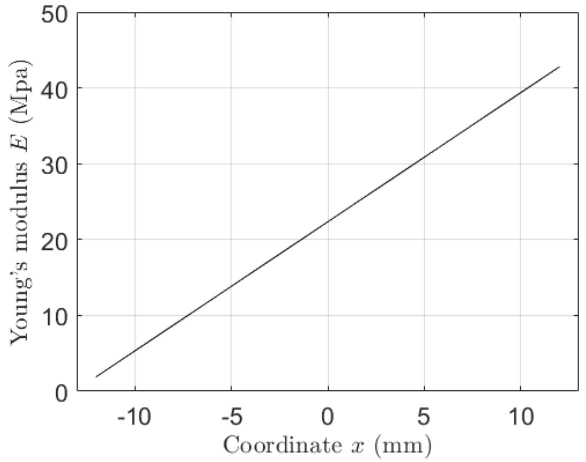
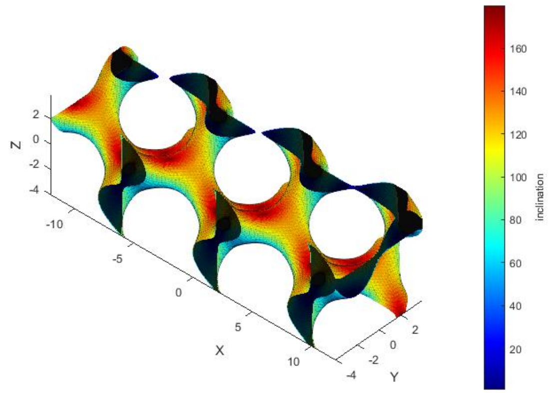
Stiffness target design



Optimized structure



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